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Plate cooling device and de-bonding station with plate cooling device

Abstract

A plate cooling device for cooling carrier plate of a wafer, includes a cooling plate and a plurality of contact pads. The cooling plate includes an upper surface; wherein the upper surface is provided with a cooling area for placing the wafer thereon. The contact pads are disposed on the upper surface. The contact pads protrude on the upper surface, and the total area of the contact pads is less than 3% of the area of the carrier plate.

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Background/Summary

BACKGROUND

Technical Field

(1) This disclosure relates to a wafer thinning process, and in particular to a plate cooling device for cooling a carrier plate in a wafer stripping operation and a de-bonding station with the plate cooling device.

Related Art

(2) The existing wafer fabrication process is moving towards the trend of thinning, but in the existing wafer thinning process, the gradual thinning of the wafer sheet has insufficient stress strength and is prone to breakage. Therefore, before thinning process, the wafer is bonded to a carrier plate, so as to enhance the mechanical strength of the wafer. And then wafer is then subjected to thinning operations such as chemical-mechanical grinding.

(3) After the thinning process, the wafer and the carrier plate are heated, to reduce the strength of the adhesive, and then the carrier plate is stripped by a vacuum pickup device and placed on a cooling platform for cooling and recycling.

(4) However, after the carrier plate is stripped from the wafer by the vacuum pickup device, the carrier plate faces the cooling platform with its bonding surface and the carrier plate has adhesive for being bonded to the wafer on the bonding surface. When the carrier plate is placed on the cooling platform, a large area of contact is made with the cooling platform by the bonding surface, so that the carrier plate is adhesively bonded to the cooling platform. When the carrier plate is transferred to other equipment after the cooling operation, the large area of adhesive creates a great adhesive force on the carrier plate, during the process of picking up and transferring the carrier plate, pickup failure may easily occur, causing the process of picking up and transferring the carrier plate to be interrupted, and the error must be eliminated manually. Meanwhile, If the adsorption force of the equipment is raised to forcefully remove the carrier plate from the cooling platform, the carrier plate will be prone to warping and deformation or rupture. Therefore, it is necessary to improve the way the carrier plate is placed on the cooling platform in order to avoid pickup failures during pickup and transfer of the carrier plate.

SUMMARY

- (5) In view of the above technical problems, this disclosure discloses a plate cooling device and a de-bonding station with plate cooling device, which can avoid errors when picking up and transferring a carrier plate from the plate cooling device.
- (6) This disclosure discloses a plate cooling device for cooling carrier plate of a wafer includes a cooling plate and a plurality of contact pads. The cooling plate includes an upper surface. The contact pads are disposed on the upper surface. The contact pads protrude on the upper surface, and the total area of the contact pads is less than 3% of the area of the carrier plate.
- (7) In one or more embodiments, the upper surface is provided with a cooling area the plurality of contact pads protrude on the upper surface.
- (8) In one or more embodiments, a front edge of the upper surface is provided with a notch.
- (9) In one or more embodiments, the upper surface is provided with at least one groove extending from the notch to a rear edge of the upper surface, and the plate cooling device further includes at least one fixing block, the at least one fixing block is provided in the groove; and the at least one fixing block is configured for a fixing member to pass through.
- (10) In one or more embodiments, the cooling plate further comprises two lateral recesses located on two opposite side edges of the upper surface, and the plate cooling device further comprises a plurality of the fixing blocks disposed on the two lateral recesses; and each of the fixing blocks is configured for a fixing member to pass through.
- (11) Based on the above-mentioned plate cooling device, this disclosure further discloses a de-bonding station for stripping a wafer from a carrier plate, and includes a base, a carrier device, the above-mentioned plate cooling device, and a carrier plate pickup device. The base is provided with two guiding pieces on two opposite edge sides. The carrier device is disposed on the base and located between the two guiding pieces. The carrier device includes a holder, a vacuum chuck, and a plurality of lifting pins. The holder is configured for the wafer and the carrier plate to be placed thereon, and the wafer faces the holder. A de-bonding section is provided on a top surface of the holder for placing the wafer and the carrier plate, and carrier plate is located above the wafer. The vacuum chuck is provided in the de-bonding section for vacuum absorbing the wafer. The lifting pins are disposed in the de-bonding section in a rising and lowering manner, and able to lower and fully embedded in the holder, or rise to push up or accept the wafer; The plate cooling device is disposed on the base and located between the two guiding pieces. The carrier plate pickup device includes a movable seat and a pickup head; wherein the movable seat is movably coupled to the two guiding pieces; and the pickup head is movably disposed on the horizontal beam for vacuum adsorbing the carrier plate, so as to strip the carrier plate from the wafer and move the carrier plate to the cooling area of the plate cooling device.
- (12) In one or more embodiments, each of the two guiding pieces is provided with a guiding slot, the movable seat comprises two pillars and a horizontal beam connecting the two pillars, the two pillars are inserted into each of the guiding slots, such that the movable seat is movably coupled to the two guiding pieces, and the pickup head is movably disposed on the horizontal beam.
- (13) In one or more embodiments, the de-bonding station further includes a first linear actuator and a second linear actuator; wherein the first linear actuator is disposed on the base and connected to one of the two pillars, the first linear actuator is configured to drive the two pillars to move along the two guiding slots; and the pickup head is connected to the horizontal beam via the second linear actuator, the second linear actuator is configured to drive the pickup head to move toward or away from the base.
- (14) In one or more embodiments, the de-bonding section is a shallow concave.
- (15) In one or more embodiments, a heater is embedded in the carrier plate to heat the wafer and the carrier plate placed in the de-bonding section.
- (16) Through the plate cooling device and the de-bonding station with plate cooling device according to this disclosure, the adhesive force generated by the adhesive on the bottom surface of

the carrier plate will not be too large, which prevents pickup failure of the carrier plate during the subsequent transfer of the carrier plate, reduces the error rate of the de-bonding process, and effectively improves the production rate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) This disclosure will become more fully understood from the detailed description given herein below for illustration only, and thus not limitative of this disclosure, wherein:
- (2) FIG. 1 is a perspective view of a de-bonding station with a plate cooling device according to an embodiment of this disclosure.
- (3) FIG. 2 is a perspective view of the plate cooling device according to the embodiment of this disclosure.
- (4) FIG. 3 is a top view of the plate cooling device according to the embodiment of this disclosure.
- (5) FIG. 4 is a lateral view of the carrier device according to the embodiment of this disclosure.
- (6) FIG. 5 is a perspective view of the carrier device according to the embodiment of this disclosure.
- (7) FIG. 6 to FIG. 11 are perspective views of the de-bonding station according to the embodiment of this disclosure, showing the stripping and cooling processes of the carrier plate.

DETAILED DESCRIPTION

- (8) Referring to FIG. 1 and FIG. 3, this disclosure discloses a plate cooling device **100** and a de-bonding station **1** with plate cooling device **100**.
- (9) As shown in FIG. 2 and FIG. 3, the plate cooling device **100** includes a cooling plate **110** and a plurality of contact pads **120**, the cooling plate **110** includes an upper surface **111**. The upper surface **111** is provided with a cooling area **110a**. The cooling area **110a** is substantially circular with a diameter slightly larger or equal to the diameter of the carrier plate **4** to be cooled. Specifically, the cooling area **110a** may be an area defined by a solid structure, such as a recess, or the cooling area **110a** may be a virtual area for setting the position of the pickup device/robot for picking up/placing the carrier plate **4**, i.e., the cooling area **110a** is an area defined only in the control codes of the pickup device/robot.
- (10) As shown in FIG. 2 and FIG. 3, The contact pads **120** are disposed on the upper surface **111**, and are located in the cooling area **110a**. The contact pads **120** protrude on the upper surface **111** and located within the cooling area **110a**.
- (11) As shown in FIG. 3, FIG. 4 and FIG. 5, The contact pads **120** are configured for the carrier plate **4** to be cooled to be placed on, and through the support of the contact pad **120**, a gap **G** is maintained between the carrier plate **4** and the upper surface **111**. That is, the carrier plate **4** is in contact only with the contact pads **120**, and the total area of contact between the carrier plate **4** and the contact pads **120** is much smaller than the area of the carrier plate **4** projected on the upper surface **111**.
- (12) As shown in FIG. 4, the carrier plate **4** is in contact only with the contact pads **120**, and there is only a small area of contact between the bottom surface of the carrier plate **4** and the contact pads **120**. Therefore, the adhesive **4a** on the bottom surface of the carrier plate **4** can only be applied to a small area, the normal adhesive force (the adhesive force in the direction normal to the upper surface **111**) is significantly reduced.
- (13) The aforementioned adhesive force can still temporarily fix the carrier plate **4** in the cooling area **110**, to prevent the carrier plate **4** from sliding on the upper surface **111**. Cooling devices can be installed inside or outside the cooling plate **110**. For example, a liquid cooling device that liquid cools the cooling plate **110** or an air-cooling device that provides a cooling airflow to the upper surface **111** to cool the carrier plate **4** directly.

(14) As shown in FIG. 3, in detail, the total area of the contact pads **120** is less than 3% of the area of the carrier plate **4**, such that the normal adhesive force (the adhesive force in the direction normal to the upper surface **111**) between the contact pads **120** and the carrier plate **4** will not be too large, so as to prevent the carrier plate pickup device **230** from being unable to pick up and move the cooled carrier plate **4** from the contact pads **120**. The configuration of the plurality of contact pads **120** may be distributed in a radial pattern from the center of the cooling area **110a** outwardly, and uniformly distributed in the cooling area **110a**. Therefore, the adhesive force is applied evenly to the bottom surface of the carrier plate **4**, constituting a uniformly distributed point adhering instead of one large area adhering.

(15) As shown in FIG. 2 and FIG. 3, in an example, a front edge **1111** of the upper surface **111** is provided with a notch **113**, the upper surface **111** is provided with one or more trenches **114**. Each of the trenches **114** extends from the notch **113** to a rear edge **1112** of the upper surface **111**. In addition, in one embodiment, the cooling plate **110** further includes two lateral recesses **115** located on two opposite side edges **113** of the upper surface **111**.

(16) As shown in FIG. 2 and FIG. 3, and the plate cooling device **100** further includes a plurality of the fixing blocks **130**, respectively disposed to the trenches **114** and the lateral recesses **115**. The fixing block **130** is configured for a fixing member (such as a screw) to pass through, so as to fix the cooling plate **110** to the de-bonding station **1** or another base.

(17) As shown in FIG. 1, the de-bonding station **1** includes a base **210**, a carrier device **220**, the plate cooling device **100**, and a carrier plate pickup device **230**.

(18) As shown in FIG. 1, The base **210** is provided with two guiding pieces **240** on two opposite edge sides **1113**. Each of the two guiding pieces **240** is provided with a guiding slot **242**. The carrier device **220** is disposed on the base **210** and located between the two guiding pieces **240**.

(19) As illustrated in FIG. 1 and FIG. 4, The carrier device **220** includes a holder **222**, a vacuum chuck **113** and a plurality of lifting pins **224**. The holder **222** is configured for placing the wafer **3** to be stripped and its carrier plate **4** thereon. A de-bonding section **222a** is provided on a top surface of the holder **222**, and the de-bonding section **222a** is a shallow concave. When the wafer **3** and the carrier plate **4** thereof are placed into the de-bonding section **222a**, the wafer **3** is placed toward the de-bonding section **222a**, so that the carrier plate **4** is positioned above the wafer **3**. A heater **225**, such as an electric heat pipe, may also be embedded in the holder **222**, so as to heat the wafer **3** and the carrier plate **4** placed in the de-bonding section **222a**. The heater **225** is not excluded from being provided outside the holder **222** for heating the carrier plate **4** directly.

(20) As shown in FIG. 4, The vacuum chuck **223** and the lifting pins **224** are disposed within the de-bonding section **222a**. The vacuum chuck **223** may include a vacuum adsorption orifice connected to a vacuum pump and an air-conducting groove connected to the vacuum adsorption orifice. The vacuum chuck **223** is configured to generate a negative pressure to adsorb the wafer **3**. The lifting pins **224** are disposed in the de-bonding section **222a** in a rising and lowering manner, and able to lower and fully embedded in the holder **222**, or rise to push up or accept the wafer **3**.

(21) As shown in FIG. 1, the plate cooling device **100** is disposed on the base **210** and located between the two guiding pieces **240**. The plate cooling device **100** is adjacent to the carrier device **220**.

(22) As shown in FIG. 1, the carrier plate pickup device **230** includes a movable seat **232** and a pickup head **234**. The movable seat **232** includes two pillars **2321** and a horizontal beam **2322** connecting the two pillars **2321**. The two pillars **2321** are inserted into each of the guiding slots **242**, such that the movable seat **232** is movably coupled to the two guiding pieces **240**, and the movable seat is able to move in a longitudinal direction relative to the base **210**. Meanwhile, the de-bonding station **1** includes first linear actuator disposed on the base **210** and connected to the movable seat **232**, particularly connected to one of the two pillars **2321**.

(23) As shown in FIG. 1, the pickup head **234** is movably disposed on horizontal beam **2322** of the movable seat **232**, and the pickup head **234** is connected to the horizontal beam **2322** via a second

linear actuator **260**. The pickup head **234** is configured for vacuum adsorbing the carrier plate **4**, the second linear drive **260** is configured to drive the pickup head **234** forward (down) toward the base **210** or away (up) from the base **210**. Meanwhile, through the first linear actuator **250** driving the two pillars **2321** to move along the guiding slots **242**, the pickup head **234** is driven to move along the longitudinal direction.

(24) Referring to FIG. **6**, which shows the de-bonding process of the carrier plate **4**. The wafer **3** is first temporarily adhered (bonded) to the carrier plate **4**. The carrier plate **4** may be, but is not limited to, a glass substrate. The carrier plate **4** is used to reinforce the mechanical strength of the wafer **3** by the backside of the wafer **3**. The wafer **3** is then placed in a thinning device to grind the surface of the wafer **3** by means of chemical-mechanical grinding, etc., to reduce the thickness of the wafer **3** and to flatten the surface of the wafer **3**. The carrier plate **4** increases mechanical strength and prevents warpage of the wafer **3** during the thinning process.

(25) As shown in FIG. **6** and FIG. **7**, The lifting pins **224** of the carrier device **220** first rise to protrude on the de-bonding section **222a**, the wafer **3** to be stripped and its carrier plate **4** are moved by a robotic arm or other transportation equipment and placed on the lifting pins **224**. And the wafer **3** faces the de-bonding section **222a** with the carrier plate **4** located above the wafer **3**.

(26) As shown in FIG. **8**, next, the lifting pins **224** lower to be fully embedded in the holder **222**, so as to place the wafer **3** to be stripped with the carrier plate **4** in the de-bonding section **222a**. The vacuum chuck **223** starts to generate a negative pressure to adsorb/fix the wafer **3**, and the heater **225** starts to heat the wafer **3** and the carrier plate **4** to weaken adhesive/bonding structure, for example, softening of adhesive between the wafer **3** and the carrier plate **4**.

(27) As shown in FIG. **9** and FIG. **10**, the second linear actuator **260** drives the pickup head **234** to lower down to contact the carrier plate **4**, and the pickup head vacuum adsorbs the carrier plate **4**. The second linear actuator **260** drives the pickup head **234** to rise to strip the carrier plate **4** from the wafer **3**.

(28) As shown in FIG. **9** and FIG. **10**, the first linear actuator **250** drives the two pillars **2321** to move along the guiding slots **242**, to move the pickup head **234** and the carrier plate **4** to the plate cooling device **100**, so as to place the carrier plate **4** on the cooling area **110a**. Through the small area of contact provided by the contact pads **120**, The adhesive force generated by the adhesive **4a** on the bottom surface of the carrier plate **4** temporarily holds the carrier plate **4** in place in the cooling area **110a** without horizontal movement.

(29) As shown in FIG. **11**, at the time, B\by means of a cooling device located inside or outside the cooling plate **110**, such as a liquid cooling device that liquid cools the cooling plate **110**, or an air-cooling device that provides a cooling airflow to the upper surface **111**, the carrier plate **4** is cooled.

(30) The cooled carrier plate **4** can be picked up by a robotic arm or other pickup device and moved to a transfer tray. Similarly, the wafer **3** in the de-bonding section **222a** can be picked up by a robotic arm or other pickup device and moved to other transfer tray. As described above, the total area of the contact pads **120** is less than 3% of the area of the carrier plate **4**, Through the small area of contact provided by the contact pads **120**, the normal adhesive force (the adhesive force in the direction normal to the upper surface **111**) between the contact pads **120** and the carrier plate **4** will not be too large, the pickup head **234** can easily pick up the carrier plate **4** without causing pickup failure or causing damage to the carrier plate **4**. The adhesion between the carrier substrate **4** and the contact pad **120** still provides sufficient shear direction adhesive force, to prevent the carrier plate **4** from moving horizontally on the cooling plate **110**.

(31) Through the plate cooling device **100** and the de-bonding station **1** with plate cooling device **100** according to this disclosure, the normal adhesive force between the contact pads **120** and the carrier plate **4** will not be too large, which prevents pickup failure of the carrier plate **4** during the subsequent transfer of the carrier plate **4**, reduces the error rate of the de-bonding process, and effectively improves the production rate.

Claims

1. A de-bonding station, for stripping a wafer from a carrier plate, comprising: a base with two guiding pieces on two opposite edge sides; a carrier device disposed on the base and located between the two guiding pieces; wherein the carrier device comprises: a holder for the wafer and the carrier plate to be placed thereon; wherein a de-bonding section is provided on a top surface of the holder for placing the wafer and the carrier plate, and carrier plate is located above the wafer, a vacuum chuck provided in the de-bonding section for vacuum absorbing the wafer; a plurality of lifting pins, disposed in the de-bonding section in a rising and lowering manner, and able to lower and fully embed in the holder, or rise to push up or accept the wafer; a plate cooling device, disposed on the base, and located between the two guiding pieces, comprising: a cooling plate including an upper surface; and a plurality of contact pads disposed on the upper surface; wherein the contact pads protrude on the upper surface, and the total area of the contact pads is less than 3% of the area of the carrier plate projected on the upper surface; and a carrier plate pickup device comprising a movable seat and a pickup head; wherein the movable seat is movably coupled to the two guiding pieces; the pickup head is movably disposed on the movable seat for vacuum adsorbing the carrier plate, so as to strip the carrier plate from the wafer and move the carrier plate to a cooling area of the plate cooling device.
 2. The de-bonding station as claimed in claim 1, wherein the upper surface is provided with a cooling area the plurality of contact pads protrude on the upper surface.
 3. The de-bonding station as claimed in claim 1, wherein a front edge of the upper surface is provided with a notch.
 4. The de-bonding station as claimed in claim 3, wherein the upper surface is provided with at least one groove extending from the notch to a rear edge of the upper surface, and the plate cooling device further includes at least one fixing block, the at least one fixing block is provided in the at least one groove; and the at least one fixing block is configured for a fixing member to pass through.
 5. The de-bonding station as claimed in claim 1, wherein the cooling plate further comprises two lateral recesses located on two opposite side edges of the upper surface, and the plate cooling device further comprises a plurality of fixing blocks disposed on the two lateral recesses; and each of the fixing blocks is configured for a fixing member to pass through.
 6. The de-bonding station as claimed in claim 1, wherein each of the two guiding pieces is provided with a guiding slot, the movable seat comprises two pillars and a horizontal beam connecting the two pillars, the two pillars are inserted into each of the guiding slots, such that the movable seat is movably coupled to the two guiding pieces, and the pickup head is movably disposed on the horizontal beam.
 7. The de-bonding station as claimed in claim 6, further comprising: a first linear actuator and a second linear actuator; wherein the first linear actuator is disposed on the base and connected to one of the two pillars, the first linear actuator is configured to drive the two pillars to move along the two guiding slots; and the pickup head is connected to the horizontal beam via the second linear actuator, the second linear actuator is configured to drive the pickup head to move toward or away from the base.
 8. The de-bonding station as claimed in claim 1, wherein the de-bonding section is a shallow concave.
 9. The de-bonding station as claimed in claim 1, wherein a heater is embedded in the carrier plate to heat the wafer and the carrier plate placed in the de-bonding section.
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